# Prakash Upadhyay

Senior Full-Stack Engineer · AI Product Engineering

Full-stack engineer with 10+ years across fintech (Gojek, Paytm Insider) and AI products (QuillBot, Inkers.ai). I ship LLM features in production, from real-time AI writing suggestions at ~14K req/s to streaming PDF-Chat, and I build my own products with agents: an agentic platform that turns issues into reviewed pull requests with a human in the loop, and 28 Claude Code skills that automate my workflow. USPTO patent holder.

## CONTACT


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## SKILLS

**AI engineering**

Claude API · Claude Code skills & plugins · Multi-agent orchestration · LLM features in production · Langfuse prompt management · SSE & WebSocket streaming · Model routing & cost budgets · Eval-driven iteration · Context retrieval · PyTorch, TensorFlow

**Frontend**

React · Next.js · React Native · TypeScript · TipTap · Tailwind CSS · Playwright · Jest

**Backend**

Node.js · NestJS · Kafka · Redis · BullMQ · REST · Event-driven design

**Data**

PostgreSQL · MySQL · MongoDB · Turso · S3, R2

**Infra & delivery**

AWS · Cloudflare Workers, Queues, Pages · Docker · Kubernetes · Nx · Turborepo · GitHub Actions · GitLab CI · Canary releases · New Relic · Sentry · ELK

## EXPERIENCE

Feb 2025 to now

### Senior Fullstack Developer, Core Tools & Growth

*QuillBot · Remote*

- Led the Grammar Checker migration to **TipTap v3** with LLM-powered real-time recommendation pipelines (grammar, writing scores, AI commands), streamed check-as-you-type over WebSockets.
- Designed and scaled rate-limited recommendation APIs for freemium access with Redis-backed usage limits at **~14K req/s** peak.
- Built **PDF-Chat** on LLMs with Langfuse prompt management, SSE streaming from the AI service and chat context persistence; extended the NestJS job service with a handler registry and pluggable provider routing.
- Ship through an Nx monorepo (30+ packages) with canary rollouts, AI code review (CodeRabbit), integration suites and A/B releases.
- Winner, QuillBot DevJam 2025 (ACE Chat, an AI writing companion); built an AI multi-brand campaign builder prototype at the Sept 2025 hackathon.

2025 to now

### Founder & Engineer, independent AI products

*Own products · Bengaluru*

- **Agentic PR Platform:** an event-driven, multi-agent system that turns GitHub issues, Slack requests and PRDs into reviewed pull requests. A risk classifier sets autonomy: safe changes ship as autonomous PRs, while medium and high-risk changes (auth, payments, infra) wait for my approval of the ADR.
- Twelve specialised agents (requirement, ADR, risk classification, planning, context discovery, test planning, coding, PR, review, QA and more) talk only through immutable, versioned artifacts with full lineage back to the issue; every run is resumable from snapshots.
- Context manifests limit what the coding agent may read; deterministic import, dependency and route graphs drive impact analysis; Claude models are routed by tier (Haiku, Sonnet, Opus with approval) under enforced cost budgets; Playwright checks every preview deploy (routes, accessibility, console, network).
- **Launchpad**, a Claude Code plugin (13 skills, 3 helper agents) that takes an idea to a branded, documented, scaffolded product in seven gated stages; used to build CertPulse, Keel and RevuPulse.

Cloudflare Workers & Queues · PostgreSQL + R2 · GitHub App · TypeScript · Claude API · Claude Code

May 2022 to Aug 2024

### Senior Fullstack Developer (Lead)

*GoTo Financial (Gojek) · Bengaluru*

- Led the frontend revamp of the **GoPay** payments app (5.6M transactions/day), cutting latency by 80% (300ms to 63ms) across one-time, recurring and PIN checkout flows.
- Built the **GoTo PIN** web SDK for PIN setup and 2FA inside merchant web and native (WebView) flows, reducing SMS OTP cost by $1.5M.
- Led scoping, risk assessment and estimation; built CI/CD on AWS S3/CloudFront with CSP-secured redirects and New Relic monitoring and alerts.

## OWN PRODUCTS

**CertPulse** Certificate infrastructure for events: import results, map them to branded templates, generate in bulk, verify by QR.
TypeScript · Turborepo · Turso

**RevuPulse** The trust layer for modern discovery.
Next.js · Nx

**noui.si** An open experiment in interfaces that form around what you are trying to do, with a voice-to-voice prototype and the open-source noui-orb web component.
React 19 · View Transitions · Claude

## PATENT

**US 10,936,288**, Voice-enabled User Interface Framework
USPTO, granted Mar 2021 (SAP Labs)

## AWARDS

Winner, QuillBot DevJam
May 2025

Winner (Logistics), SAP Global P&I IoT Challenge
Sep 2016

Winner, SAP sudoCODE 3.0
Jun 2016

Winner, SAP CodeForCharity
May 2016

Podium, SAP ECS ML Hackathon
Oct 2017

## EDUCATION

**B.E. Computer Science**
Vivekanand Education Society's Institute of Technology, Mumbai
2010 to 2014

## CERTIFICATIONS

Software Architecture: From Developer to Architect

Learning Kubernetes

Deep learning foundations (TSAI EVA): CNNs, VAEs, computer vision, NLP

## EXPERIENCE, CONTINUED

Apr 2021 to Apr 2022

### Senior Software Developer

*Paytm Insider · Mumbai (Remote)*

- Architected the live video platform for insider.in events (**50+ concurrent hosts, 3K+ viewers**) on 100ms, with live chat, polls, Q&A and leaderboards.
- Built the video tile rendering engine from scratch to minimise re-renders on low-bandwidth phones; LCP under 0.5s.
- Implemented role-based streaming capabilities for super admins, hosts and audiences.

Apr 2019 to Jul 2020

### Fullstack Developer

*Inkers.ai · Bengaluru*

- Built **computer-vision retail analytics** (real-time engagement tracking, heat maps, reports) and NLP search over aircraft repair concession documents.
- Built admin dashboards with RBAC for health, retail, aerospace and co-working clients; materialised OLAP queries and client caching cut page load by 40%.

Sep 2018 to Feb 2019

### Freelance project

*TestPod, an ed-tech SaaS · Bengaluru*

- Built TestPod, an ed-tech SaaS product, as a freelance project.

Jun 2015 to Jun 2018

### Software Developer

*SAP Labs · Bengaluru*

- Invented a **voice-enabled UI framework** for enterprise web apps, granted as US patent 10,936,288.
- As a fellow in SAP's Machine Learning team (Singapore), built TensorFlow models for gas-meter reading and apparel classification, deployed on EC2 with Docker.
- Built a Go CLI for ML model management and training pipelines with Spring Boot APIs, load-tested with JMeter.

Jul 2014 to May 2015

### Associate Software Engineer

*Accenture (Avanade) · Mumbai*

- Automated the test suite for 9-1-1 emergency software with CodedUI, taking coverage to 90%.

## OPEN WORK

**HDR brightness preference** Proposed a CSS user preference for sudden HDR brightness spikes.
w3c/csswg-drafts#14616 · Oct 2026

**28 Claude Code skills** across 5 plugins: content, video (Remotion, ElevenLabs, Whisper), design and daily ops.